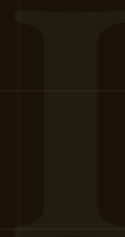


VETERINARY MICROBIOLOGY • PART 4

Veterinary Microbiology

Complete exam-ready notes covering all major veterinary viral pathogens — Paramyxoviruses, Orthomyxoviruses, Rhabdoviruses, Picornaviruses, Togaviruses, Flaviviruses, Coronaviruses, Reoviruses, Birnaviruses, Retroviruses, Poxviruses, Asfarviruses, Herpesviruses, Adenoviruses, Parvoviruses, Papovaviruses, and Circoviruses. Based on ICAR eCourses.

Subjects	Parts	Chapters	Edition
1	1	31	Extendable



01	Introduction to Systematic Veterinary Virology	02	Newcastle Disease
03	Rinderpest	04	Peste des Petits Ruminants (PPR)
05	Canine Distemper	06	Avian Influenza
07	Equine Influenza	08	Swine Influenza
09	Rabies	10	Bovine Ephemeral Fever
11	Foot and Mouth Disease	12	Duck Viral Hepatitis
13	Equine Encephalomyelitis	14	Classical Swine Fever
15	Bovine Viral Diarrhoea & Mucosal Disease	16	Avian Infectious Bronchitis
17	Transmissible Gastroenteritis (TGE)	18	Bluetongue
19	African Horse Sickness	20	Infectious Bursal Disease (Gumboro)
21	Equine Infectious Anaemia	22	Sheep Pox and Goat Pox
23	Fowl Pox	24	Contagious Ecthyma (Orf)
25	Other Pox Diseases	26	African Swine Fever
27	Herpesviruses	28	Adenoviruses
29	Parvoviruses	30	Papovaviruses
31	Circoviruses		

◆ PART 4

VETERINARY MICROBIOLOGY

Systematic Veterinary Virology

ICAR eCourses

Introduction to Systematic Veterinary Virology

Systematic Veterinary Virology deals with the study of individual viruses and the diseases they cause in domestic and wild animals. This module covers major veterinary viral pathogens grouped by family, with emphasis on classification, morphology, pathogenesis, diagnosis, and control.

SCOPE OF SYSTEMATIC VIROLOGY

The course covers RNA viruses (Paramyxoviridae, Orthomyxoviridae, Rhabdoviridae, Picornaviridae, Togaviridae, Flaviviridae, Coronaviridae, Reoviridae, Birnaviridae, Retroviridae) and DNA viruses (Poxviridae, Asfarviridae, Herpesviridae, Adenoviridae, Parvoviridae, Papovaviridae, Circoviridae).

BALTIMORE CLASSIFICATION — QUICK REFERENCE

GROUP	NUCLEIC ACID	KEY FAMILIES
Group I	dsDNA	Herpesviridae, Poxviridae, Adenoviridae, Asfarviridae
Group II	ssDNA	Parvoviridae, Circoviridae, Papovaviridae
Group III	dsRNA	Reoviridae, Birnaviridae
Group IV	+ss RNA	Picornaviridae, Togaviridae, Flaviviridae, Coronaviridae
Group V	−ss RNA	Paramyxoviridae, Orthomyxoviridae, Rhabdoviridae
Group VI	ssRNA-RT	Retroviridae

OIE LIST A DISEASES (NOTIFIABLE)

Several viral diseases in this course are OIE List A notifiable diseases including **Foot and Mouth Disease**, **Newcastle Disease**, **Avian Influenza (HPAI)**, **Rinderpest**, **Classical Swine Fever**, **African Swine Fever**, and **African Horse Sickness**. These are transboundary animal diseases (TADs) with potential for rapid spread and serious socioeconomic impact.

KEY CONCEPT — PATHOGENICITY DETERMINANTS

For all systematic viruses, pathogenicity depends on: (1) viral factors — surface proteins enabling attachment and entry; (2) host factors — species susceptibility, age, immune status; (3) environmental factors — stress, co-infections.

Newcastle Disease

Newcastle Disease (ND) is caused by **Avian Paramyxovirus type 1 (APMV-1)**. It is one of the most severe viral diseases of birds, causing heavy economic losses in poultry worldwide and is endemic in most countries.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V — Negative sense RNA viruses
Order	Mononegavirales
Family	Paramyxoviridae
Subfamily	Paramyxovirinae
Genus	Avulavirus
Species	Avian Paramyxovirus 1 (APMV-1)

There are nine different species under the genus *Avulavirus* (APMV-1 to APMV-9). APMV-1 causes Newcastle disease. APMV-1 cross-reacts with APMV-2 and APMV-3.

MORPHOLOGY

APMV-1 is a **single-stranded, non-segmented, negative-sense RNA virus**. They are enveloped and ether-sensitive, with helical symmetry. Surface projections are the HN protein. They measure **150–300 nm**.

The virus contains **6 genes** encoding 6 major structural proteins:

- **NP** — Nucleoprotein
- **P** — Phosphoprotein
- **M** — Matrix protein
- **F** — Fusion protein (critical for pathogenesis)

- **HN** — Hemagglutinin-neuraminidase (attachment + release)
- **L** — RNA-dependent RNA polymerase

Two non-structural proteins **V** and **W** are also present. The RNP complex = RNA + NP + P + L.

SYNONYMS

Pseudo fowlpest, **Ranikhet disease** (India), Vellaikalichal (vernacular Tamil).

STRAINS AND PATHOTYPES

PATHOTYPE	STRAINS	DISEASE TYPE
Velogenic	Hertz 33, Texas GB	Most virulent; systemic or neurological
Mesogenic	K, R2B	Intermediate virulence
Lentogenic	F, LaSota, Ulster 2C	Mild respiratory; used as live vaccines
Avirulent	V4	No disease

Five pathotypes based on predominant signs: **Viscerotropic velogenic** (Doyle's form — haemorrhagic intestinal lesions), **Neurotropic velogenic** (Beach's form — respiratory + nervous signs), **Mesogenic** (Beaudette's form — respiratory with some mortality), **Lentogenic** (Hitchner's form — mild respiratory), **Asymptomatic enteric**.

OIE LISTING & RISK GROUP

OIE **List A** infection (ND and HPAI are the only two poultry diseases in List A). Risk group: **Group II** Animal Pathogen.

HA PROPERTY

APMV-1 agglutinates chicken and human type O erythrocytes. After ~30 minutes, agglutination is lost due to receptor cleavage by neuraminidase — this phenomenon is called **elution**.

CULTIVATION

Readily cultivated in **embryonated eggs** via allantoic cavity route. 9–11 day old embryos preferred. Virulent strains kill embryo in 24 h with haemorrhages. Virus presence detected by HA test on allantoic fluid. Also grows in cell cultures (Vero, chicken embryo fibroblasts); CPE = multinucleated giant cells (syncytia).

RESISTANCE

APMV-1 is relatively heat-sensitive. Inactivated at 56°C for 3 hours or 60°C for 30 minutes. Stable at pH 2–10. Susceptible to common disinfectants (quaternary ammonium, phenols, oxidizing agents, formalin). Survives in litter and fomites for weeks.

PATHOGENESIS

Transmission: aerosol inhalation, direct contact, fomites, contaminated feed/water. Entry via respiratory/digestive tract → local replication → viremia → neurological and visceral involvement.

Incubation: **2–15 days**.

CLINICAL SIGNS

Respiratory (gasping, coughing), digestive (green diarrhoea), nervous (paralysis, torticollis, circling), reproductive (cessation of egg production, soft-shelled eggs). Sudden high mortality in velogenic strains.

DIAGNOSIS

Field: Clinical signs (nervous + respiratory + digestive). **Lab:** Virus isolation in embryonated eggs (HA of allantoic fluid) + HI test for serotyping. PCR for rapid detection. Intracerebral pathogenicity index (ICPI) in day-old chicks: $\text{ICPI} \geq 0.7$ = velogenic.

ICPI — EXAM POINT

Intracerebral Pathogenicity Index (ICPI): ≥ 0.7 = notifiable ND. Lentogenic strains have ICPI near 0.

VACCINES

Live attenuated vaccines (LaSota, F, Clone-30, Hitchner B1) and inactivated oil-emulsion vaccines. Live vaccines given via drinking water, eye drop or spray. Killed vaccines given by injection for long-lasting

immunity.

| CONTROL

Strict biosecurity: bird-proofing, isolation of flocks, disinfection of equipment, restriction on movements. Regular vaccination + antibody monitoring. Stamping-out policy in disease-free countries. ND is a reportable disease in most countries.

| PUBLIC HEALTH

Humans may develop mild conjunctivitis after exposure. No transmission through poultry products known. No systemic disease in humans. Occurs mainly in lab workers and vaccinating crews.

Rinderpest

Rinderpest (cattle plague) is an acute viral disease of domestic cattle, buffaloes, and yaks caused by **Rinderpest virus (RPV)**. Characterized by high morbidity and mortality, it was the first animal disease to be globally eradicated in 2011.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V — Negative sense RNA
Order	Mononegavirales
Family	Paramyxoviridae
Genus	Morbillivirus
Species	Rinderpest virus

MORPHOLOGY

Pleomorphic, 150–300 nm, enveloped, helical nucleocapsid, ssRNA negative sense, non-segmented. Six major proteins: **N, P, M, F, H, L** + two non-structural proteins **C/V**. The **H (hemagglutinin)** and **F (fusion)** proteins are critical for infection — H for attachment, F for membrane fusion. RPV does *not* have neuraminidase activity.

STRAINS

Three lineages (Asian, African, mild). A single serotype. Closely related to measles virus (human), canine distemper virus, PPR virus, and phocine distemper virus.

CULTIVATION

Cultivated in calf kidney cells, African green monkey cells (Vero), B95a (marmoset lymphoblastoid cells). CPE: small foci of rounded cells → stellate cells → multinucleate syncytia (by day 9–10). Grows

in chorioallantoic membranes of embryonated eggs. Rabbits are the lab animal of choice.

| PATHOGENESIS

Transmission: aerosol, direct contact. Virus replicates in lymphoid tissue → viremia → fever, oral erosions, diarrhoea, lymphoid necrosis. Incubation: 3–15 days. Morbidity and mortality can reach 100% in naive populations.

| CLINICAL SIGNS

Fever (40–41°C), oral erosions/necrotic lesions, profuse mucopurulent nasal and ocular discharge, severe haemorrhagic diarrhoea, dehydration, death. Characteristic **oral erosions** = cheesy necrotic lesions on gums, inner lips, tongue.

| DIAGNOSIS

Field: High morbidity + fever + oral erosions + haemorrhagic diarrhoea in cattle. **Lab:** Virus isolation, AGID, CFT, ELISA (competition ELISA — c-ELISA is gold standard), FAT on lymph node impression smears. PCR for rapid diagnosis.

ERADICATION OF RINDERPEST

Rinderpest was officially declared eradicated in June 2011 by FAO and OIE — the first animal disease to be globally eradicated and only the second disease ever eradicated (after smallpox in humans). The Global Rinderpest Eradication Programme (GREP) was operated by FAO under EMPRES.

| VACCINES

Plowright attenuated Vero cell-adapted vaccine (RBOK strain) — the vaccine that achieved global eradication. Gives lifelong immunity with a single dose. Thermostable variants developed for field use.

| PUBLIC HEALTH

No rinderpest infection has been reported in humans.

Peste des Petits Ruminants (PPR)

Peste des Petits Ruminants (PPR) — also called **goat plague** — is an acute or subacute viral disease of goats and sheep caused by PPRV (Morbillivirus genus). First described in 1942 in Côte d'Ivoire, West Africa. Closely related to rinderpest, canine distemper, and measles viruses.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V
Order	Mononegavirales
Family	Paramyxoviridae
Genus	Morbillivirus
Species	PPR virus (PPRV)

MORPHOLOGY

Pleomorphic, ~150 nm, enveloped, helical symmetry, ssRNA negative sense. Goats more severely affected than sheep. OIE **List A** — Class II Animal Pathogen.

STRAINS

Four lineage groups — 3 in Africa, 1 in Asia (one African group also prevalent in Asia). Indian vaccine strain: **AR-87 (Arasur strain)** developed at Madras Veterinary College. Other South Indian isolates: Coimbatore (CBE) and Poondi (92).

RESISTANCE

Heat-sensitive — inactivated at 56°C. Stable at pH 5.8–10.0. Sensitive to ether and detergents. Survives several hours outside the host.

HOSTS & DISTRIBUTION

Primary hosts: sheep and goats (goats more severely affected). Cattle, buffaloes, camels, pigs are susceptible but don't show clinical signs. Wild ungulates affected. Distribution: Sub-Saharan Africa, Indian subcontinent, Middle East.

TRANSMISSION

Not very contagious; requires close contact. Aerosol (sneezing/coughing) is the main route. Ocular, nasal, oral secretions and faeces are virus sources. Fomites also implicated.

CLINICAL SIGNS

Fever (40–41°C), erosive stomatitis (necrotic lesions in mouth), conjunctivitis, gastroenteritis (profuse foul-smelling diarrhoea), pneumonia. Mortality can reach 70–80% in severe outbreaks.

DIAGNOSIS

AGID, CFT, virus neutralization, ELISA (c-ELISA using anti-N antibody). PCR. Virus isolation in Vero cells; CPE similar to rinderpest — syncytia formation.

VACCINES

Tissue culture rinderpest vaccine (TCRV) gives 12 months protection but interferes with rinderpest serosurveillance. Homologous attenuated PPR vaccine (AR-87 strain) used in southern India. Nigeria 75/1 strain used as standard vaccine in Africa.

TREATMENT & CONTROL

No specific treatment. Oxytetracycline and chlortetracycline reduce secondary infections. Control: quarantine, stamping out, ring vaccination, movement restrictions. Eradication methods same as rinderpest.

PUBLIC HEALTH

PPR does not infect humans — no public health concern.

Canine Distemper

Canine Distemper is a contagious, incurable, multi-systemic, often fatal viral disease of dogs and other carnivores caused by **Canine Distemper Virus (CDV)**. It is one of the most important infectious diseases of dogs worldwide.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V — Negative sense RNA
Order	Mononegavirales
Family	Paramyxoviridae
Genus	Morbillivirus
Species	Canine distemper virus

MORPHOLOGY

Enveloped, pleomorphic, 150–300 nm, ssRNA negative sense, helical nucleocapsid. Six structural proteins: N, P, M, F, H, L. No neuraminidase. Related to human measles, rinderpest, and PPR viruses.

HOSTS

Dogs (Canidae), ferrets, mink (Mustelidae), raccoons (Procyonidae), and several large felids. Phocine distemper virus (PDV) infects seals and is closely related.

PATHOGENESIS

Transmission: respiratory aerosols. Virus replicates in lymphoid tissue → viremia → spreads to epithelial and nervous tissue. Characteristic of immunosuppression followed by secondary bacterial infections.

CLINICAL SIGNS — THE THREE FORMS

Respiratory form: coughing, nasal discharge, pneumonia. **Enteric form:** vomiting, diarrhoea.

Nervous form: chorea (muscle twitching), seizures, ataxia, hyperkeratosis of paw pads and nose (**hard pad disease**). Old Dog Encephalitis (ODE) = chronic progressive neurological disease in older dogs due to persistent CDV infection.

HARD PAD DISEASE

Hyperkeratosis of the nose and footpads is pathognomonic for canine distemper. Also called "hard pad disease."

INCLUSION BODIES

Intranuclear and intracytoplasmic **eosinophilic inclusion bodies** in epithelial cells and neurons (Lentz bodies).

DIAGNOSIS

Clinical signs + inclusion bodies on conjunctival smears (FAT). Serology (VN test, ELISA). PCR. Virus isolation in Vero cells (CPE: syncytia).

VACCINES

Live attenuated vaccines (Snyder Hill, Lederle strains) and killed vaccines. MLV vaccines provide excellent immunity. Combination vaccines (DHPP) are standard. Maternal antibodies can interfere — primary vaccination at 8–10 weeks, booster at 12–16 weeks.

Avian Influenza

Avian Influenza (AI) — also called **fowl plague** — is caused by Type A influenza viruses. Highly Pathogenic Avian Influenza (HPAI) is an OIE List A disease causing up to 100% mortality in affected poultry flocks.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V — Negative sense RNA
Order	Articulavirales
Family	Orthomyxoviridae
Genus	Influenzavirus A

MORPHOLOGY

Enveloped, spherical to pleomorphic, **80–120 nm**. Helical nucleocapsid. ssRNA negative sense, **segmented genome – 8 RNA segments**. This segmentation enables **antigenic shift** (reassortment).

MAJOR SURFACE PROTEINS

Hemagglutinin (HA): 16 subtypes (H1–H16). Binds sialic acid receptors; enables cell entry; main antigen for antibody neutralization.

Neuraminidase (NA): 9 subtypes (N1–N9). Cleaves sialic acid to release progeny virions; enables spread.

16 HA × 9 NA = theoretically 144 subtypes possible.

HPAI VS LPAI

HPAI (Highly Pathogenic): H5 and H7 subtypes. IVPI ≥ 1.2 or multiple basic amino acids at HA cleavage site. Causes systemic infection, hemorrhages, near 100% mortality. **LPAI**: Mild respiratory/enteric disease. H5N1 (Asian lineage) is the most significant HPAI strain globally.

ANTIGENIC VARIATION

Antigenic Drift: Gradual mutations in HA/NA genes (point mutations). **Antigenic Shift**: Reassortment of gene segments from two different influenza strains co-infecting one cell — creates new subtype. Shift is responsible for pandemics.

HOST RANGE

Wild aquatic birds (waterfowl) = natural reservoir. Chickens, turkeys, ducks, geese, all poultry. Occasionally pigs, horses, marine mammals, and humans (H5N1, H7N9). Pigs serve as **mixing vessel** for avian and human strains.

PATHOGENESIS OF HPAI

Entry via respiratory/oral routes → systemic viraemia → multiorgan damage. Basic amino acid cleavage site in HA allows cleavage by ubiquitous proteases in all tissues → severe systemic disease. Peracute death with few signs. Cutaneous oedema, haemorrhages, necrosis in multiple organs.

CLINICAL SIGNS

HPAI: Sudden death, severe depression, cyanosis of comb/wattles, haemorrhages, cessation of egg production, neurological signs. LPAI: Mild respiratory signs, drop in egg production.

DIAGNOSIS

Virus isolation (embryonated eggs — allantoic cavity). HA and HI tests for serotyping. ELISA, FAT. RT-PCR for rapid subtype identification. Intravenous Pathogenicity Index (IVPI) in chickens for pathotype classification.

| VACCINES & CONTROL

Inactivated whole-virus vaccines. Recombinant vaccines (fowlpox-H5 vectored). Emergency vaccination (ring vaccination). Stamping-out + disinfection in disease-free countries. AI is notifiable to OIE; HPAI is List A.

| PUBLIC HEALTH — ZOONOSIS

H5N1 HPAI: human cases with high fatality rate (~60%). H7N9: human cases in China. Transmission mainly from direct contact with infected birds. No sustained human-to-human transmission of current strains. Pandemic risk if reassortment creates efficient human-transmissible strain.

Equine Influenza

Equine Influenza (EI) is a highly contagious acute respiratory disease of horses caused by **Equine Influenza A virus** (subtypes H3N8 and H7N7). It is the most important respiratory disease of horses worldwide.

CLASSIFICATION

Baltimore Group V. Family Orthomyxoviridae, Genus Influenzavirus A. Two subtypes: **H7N7** (Equi-1; antigenically stable, rarely isolated now) and **H3N8** (Equi-2; currently dominant, continuously evolving).

MORPHOLOGY & PROPERTIES

Similar to other influenza A viruses — enveloped, 80–120 nm, 8-segment ssRNA (–sense). HA and NA are major surface antigens. Antigenic drift occurs continuously in H3N8.

HOSTS & TRANSMISSION

Horses, donkeys, mules. Highly contagious — aerosol spread. Incubation 1–3 days. Outbreak in naïve populations can be explosive.

CLINICAL SIGNS

Sudden onset fever (40–41°C), harsh dry cough (characteristic), bilateral serous/mucopurulent nasal discharge, limb oedema, myalgia. Recovery in 2–3 weeks if uncomplicated. Secondary bacterial pneumonia is the main complication.

DIAGNOSIS

Virus isolation (embryonated eggs or MDCK cells), HI test, ELISA, RT-PCR. Paired serology (4-fold rise in HI titre).

VACCINES

Inactivated bivalent vaccines (H7N7 + H3N8). Annual vaccination required due to antigenic drift.

Intranasal MLV vaccine available. Canary pox-vectored recombinant vaccine (Recombitek) available.

Swine Influenza

Swine Influenza (SI) is an acute, highly contagious respiratory disease of pigs caused by **Swine Influenza A viruses**. Important because pigs serve as a **mixing vessel** for reassortment of avian and human influenza viruses.

CLASSIFICATION

Baltimore Group V. Family Orthomyxoviridae, Influenzavirus A. Major subtypes in pigs: **H1N1**, **H3N2**, **H1N2**. The 2009 pandemic H1N1 (pH1N1) was a reassortant swine-origin strain.

MORPHOLOGY & PROPERTIES

Same structure as other influenza A viruses — enveloped, 80–120 nm, 8-segment ssRNA (–sense). Classical swine H1N1 is antigenically distinct from human H1N1.

THE MIXING VESSEL CONCEPT

Pig trachea has **both α -2,3 (avian-type) and α -2,6 (human-type) sialic acid receptors**. This allows simultaneous infection by avian and human influenza strains → reassortment of gene segments → novel subtypes with pandemic potential.

HOSTS & TRANSMISSION

Pigs primarily. Can infect humans (zoonosis). Aerosol/droplet transmission. Outbreaks typically in autumn and winter.

CLINICAL SIGNS

Sudden onset fever, coughing, labored breathing, nasal discharge, anorexia, prostration. High morbidity, low mortality. Secondary bacterial pneumonia (*Pasteurella*, *Haemophilus*) increases severity.

PATHOLOGY

Bronchointerstitial pneumonia. Lung consolidation in cranioventral portions. Histologically: necrotizing bronchiolitis, inflammatory infiltrates.

DIAGNOSIS

Virus isolation (MDCK cells or embryonated eggs), RT-PCR, ELISA, HI test. FAT on lung tissue.

VACCINES & CONTROL

Inactivated whole-virus vaccines (H1N1, H3N2 components). Autogenous vaccines when new variant emerges. Biosecurity, all-in/all-out management. Work with infectious material in Class II biological safety cabinet.

Rabies

Rabies is a fatal neurological disease caused by **Rabies virus (RABV)**. It is virtually 100% fatal once symptoms appear. It is a major zoonosis causing ~59,000 human deaths annually, mostly in Asia and Africa.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group V — Negative sense RNA
Order	Mononegavirales
Family	Rhabdoviridae
Genus	Lyssavirus
Species	Rabies lyssavirus

MORPHOLOGY

Bullet-shaped — approximately 180 nm long × 75 nm wide. Enveloped, ether-sensitive. ssRNA negative sense, non-segmented. Genome encodes **5 proteins: N, P, M, G, L**. The **G (glycoprotein)** forms ~400 trimeric spikes — mediates attachment, entry, and is the major target of virus-neutralizing antibodies.

STREET VIRUS VS FIXED VIRUS

Street virus: Virulent virus from rabid animals in nature — long variable incubation, produces Negri bodies, neurotropic. **Fixed virus**: Adapted by serial passages in laboratory animals (rabbits) → shortened incubation, faster replication, loses Negri body formation, less pathogenic, used for vaccine production.

Important strains: Flury (HEP, LEP), SAD (Street-Alabama-Dufferin), Vnukovo, Kelev.

CULTIVATION

BHK-21 (baby hamster kidney) cells and neuroblastoma cell lines (CCL-131). Grows in suckling mouse brain (SMB). Demonstrated by FAT (fluorescent antibody test) in infected cells.

NEGRI BODIES

Intracytoplasmic eosinophilic inclusion bodies in neurons, especially in **Purkinje cells of cerebellum** and pyramidal cells of hippocampus (Ammon's horn). **Magenta coloured** with small dark-blue interior granules on Mann's/Sellers staining. Pathognomonic for rabies.

NEGRI BODIES — EXAM ESSENTIAL

Negri bodies are found in the cytoplasm of neurons. Best seen in: (1) Hippocampus (Ammon's horn), (2) Cerebellum (Purkinje cells), (3) Cerebral cortex. Stained by Sellers, Mann's, or Giemsa stain — appear magenta/red with blue interior granules. Pathognomonic for rabies.

PATHOGENESIS

Virus enters via bite wound → replicates in muscle at bite site → binds nicotinic acetylcholine receptors at neuromuscular junctions → ascends centripetally via peripheral nerves to CNS → replicates in brain → descends centrifugally to salivary glands, cornea, skin. **Centripetal then centrifugal spread.**

CLINICAL SIGNS

Prodromal phase: Behaviour change, anxiety (2–3 days). **Excitative (furious) phase:** Aggression, delirium, hydrophobia in humans, "mad dog syndrome" in dogs. **Paralytic (dumb) phase:** Progressive paralysis, coma, death. Death follows onset of symptoms by 7–10 days.

DIAGNOSIS

Gold standard: FAT (Fluorescent Antibody Test) on brain impression smears — sensitivity ~98%. Virus isolation in neuroblastoma cells or suckling mice. Direct rapid immunohistochemical test (DRIT). Detection of Negri bodies (Sellers stain). Ante-mortem: saliva, CSF, nuchal skin biopsy, corneal smears for FAT/RT-PCR.

VACCINES

TYPE	EXAMPLES	NOTES
Neural vaccines (obsolete)	Semple (infected rabbit brain), Fuenzalida (suckling mouse brain)	Risk of neuromuscular reactions
Cell culture vaccines	HDCV, PCEC, PVRV (Verorab)	Safe, immunogenic — gold standard
Animal vaccines	ERA, Flury HEP, SAD-Bern	Oral bait vaccines for wildlife

Semple vaccine: Neuromuscular reactions in 4.6/1000 vaccinated, 3.13% fatality — now replaced by cell culture vaccines in most countries. Post-exposure prophylaxis (PEP) = local wound cleaning + passive immunization (RIG) + vaccine series.

CONTROL

Dog vaccination (>70% population for herd immunity). Stray dog control. Pre-exposure vaccination for high-risk individuals (veterinarians, animal handlers). PEP immediately after exposure. Wildlife oral vaccination using bait vaccines.

Bovine Ephemeral Fever

Bovine Ephemeral Fever (BEF) — also called **Three-day sickness** — is an arthropod-borne, non-contagious, febrile disease of cattle and water buffaloes. Caused by **Bovine Ephemeral Fever virus (BEFV)**.

CLASSIFICATION

Baltimore Group V. Family Rhabdoviridae, Genus Ephemerovirus. Bullet-shaped, enveloped, ssRNA negative sense. Related to rabies but in different genus.

VECTOR & TRANSMISSION

Transmitted by biting arthropods — **Culicoides** midges and mosquitoes (Culicine and Anopheline). Not directly contagious between animals. Occurs in seasonal outbreaks following wet weather (breeding of insect vectors). Distribution: Africa, Middle East, Asia, Australia.

CLINICAL SIGNS

Sudden high fever (40–41°C), loss of milk production, stiffness/lameness, reluctance to move, recumbency, salivation, nasal discharge. Most animals recover in 1–3 days (**hence "three-day sickness"**). Mortality is low but milk loss and temporary infertility cause economic losses. Recumbent animals may develop aspiration pneumonia.

PATHOLOGY

Polyserositis, fibrinous polyarthritis, myositis. Inflammatory changes in synovial membranes, periarticular tissues, and muscles.

DIAGNOSIS

Clinical diagnosis: characteristic 3-day fever with stiffness in enzootic region. Virus isolation, serology (VN test), ELISA.

VACCINES

Live attenuated, killed, and subunit vaccines used in some parts of world. Live vaccine: ≥ 12 months protection after 2 doses. Inactivated vaccine: 6 months protection. Vaccination from 6 months of age, annual revaccination.

TREATMENT

NSAIDs (flunixin, phenylbutazone) to reduce fever and inflammation. Calcium borogluconate for hypocalcaemia. Most animals recover spontaneously. Supportive care for recumbent animals.

Foot and Mouth Disease

Foot and Mouth Disease (FMD) is the most economically important livestock disease worldwide. Caused by **Foot and Mouth Disease Virus (FMDV)**, it is highly contagious and affects cloven-hoofed animals. OIE **List A** disease.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group IV — Positive sense RNA
Family	Picornaviridae
Genus	Aphthovirus
Species	Foot-and-mouth disease virus

MORPHOLOGY

Non-enveloped, icosahedral, **25–30 nm**. ssRNA positive sense. No lipid envelope — resistant to ether/detergents. Capsid proteins: VP1, VP2, VP3, VP4. **VP1** contains the main antigenic sites (RGD motif — integrin binding).

SEROTYPES

7 serotypes: **O, A, C, Asia 1, SAT 1, SAT 2, SAT 3**. No cross-immunity between serotypes. Over 60 subtypes within serotypes. Serotype O is the most widespread globally. Serotype C is nearly extinct. India: mainly O, A, Asia 1 (SAT strains not endemic in India).

7 FMDV SEROTYPES — MNEMONICS

O A C (Old World/Europe) + **Asia 1** + **SAT 1, 2, 3** (Southern African Territories). 7 serotypes, no cross-protection.

RESISTANCE

pH sensitive: inactivated at pH <6.5 or >11. Survives in milk/dairy products (pH 4.6 at 70°C for 15 seconds). Heat: inactivated at 60°C for 30 minutes. Resistant to ether (non-enveloped). Survives in bone marrow of frozen carcasses.

REPLICATION STEPS (TIMELINE)

0h: Virus attachment → 0–3h: RNA release + translation → 3–4h: Plasma membrane permeabilization → 4–6h: Virus assembly in cytoplasm → 6–10h: Cell lysis and release.

HOSTS

All cloven-hoofed animals: cattle, buffaloes, pigs, sheep, goats, deer, camels. Pigs are **amplifying hosts** (excrete large quantities of virus). Sheep are often subclinical. Hedgehogs also susceptible. Horses are NOT susceptible.

TRANSMISSION

Most contagious animal disease known. Aerosol (main route), direct contact, fomites, contaminated feed/equipment. Pigs can generate large aerosol clouds. Virus persists in frozen carcasses, litter, hay. Incubation 2–14 days.

CLINICAL SIGNS

Fever (40–41°C), vesicles (blisters) on tongue, dental pad, gums, nares, coronary band, interdigital space, teats. Vesicles rupture → erosions → lameness, excessive salivation, drop in milk production. Low mortality in adults, higher in young animals. Cardiac death in piglets ("tiger heart" — pale myocardial streaks).

CARRIER STATE

Cattle, sheep, goats can become carriers — virus persists in pharyngeal region for up to **3.5 years (cattle), 9 months (sheep)**. Pigs do NOT become carriers. Carriers are important for maintaining virus in populations.

DIAGNOSIS

Probe of choice: Epithelium from ruptured vesicles. Tests: Complement fixation test (CFT), ELISA, Virus isolation in BHK-21 cells or primary calf thyroid cells, RT-PCR. NSP ELISA differentiates vaccinated from infected animals (non-structural protein antibodies).

VACCINES

Inactivated (killed) vaccines from BHK-21 cells, adjuvanted with aluminium hydroxide + saponin or oil emulsion. Trivalent or multivalent vaccines combining 2–3 serotypes. Requires cold chain. Protection for 4–6 months — biannual vaccination in endemic areas. Emergency vaccination ring policy in outbreaks.

DIFFERENTIAL DIAGNOSIS

Vesicular Stomatitis (VS), Swine Vesicular Disease (SVD), Vesicular Exanthema of Swine (VES) — all cause similar vesicular lesions. Differentiation by virus isolation and serology.

Duck Viral Hepatitis

Duck Viral Hepatitis (DVH) is an acute, highly fatal disease of young ducklings caused by **Duck Hepatitis Virus (DHV)**. Three distinct serotypes (I, II, III) exist; type I is most widespread and economically important.

CLASSIFICATION

DHV type I: Baltimore Group IV. Family Picornaviridae, Genus Avihepatovirus. Small, non-enveloped, icosahedral, 20–40 nm, ssRNA positive sense.

HOSTS & SUSCEPTIBILITY

Primarily affects ducklings under 3 weeks of age. Most severe in birds under 1 week. Older ducks (>5 weeks) are naturally resistant. Geese and turkey poults may also be susceptible to some strains.

TRANSMISSION

Highly contagious. Faecal-oral route. Virus persists in litter and environment for weeks.

CLINICAL SIGNS

Peracute course — sudden death with opisthotonus (spasmodic backward curving of neck and body). Ducklings run in circles, fall on their sides, paddle their legs, and die. Very high mortality (up to 95% in 1-week-old ducklings). Postmortem: swollen, haemorrhagic liver with petechiae and ecchymoses.

OPISTHOTONUS IN DVH

Spasmodic backward arching of neck = opisthotonus. Pathognomonic finding in dead ducklings with DVH. Liver shows haemorrhagic hepatitis on post-mortem.

DIAGNOSIS

Gross lesions + age of affected birds. Virus isolation in embryonated duck eggs (death + hepatitis in embryo) or 1-day-old susceptible ducklings. Serology: VN test, ELISA.

VACCINES & CONTROL

Live attenuated vaccine given to breeders — maternal antibodies protect ducklings for 2–3 weeks.

Emergency treatment: antiserum from immune ducks reduces mortality. Good sanitation and all-in/all-out management.

Equine Encephalomyelitis

Equine Encephalomyelitis refers to a group of arthropod-borne viral encephalitides of horses caused by alphaviruses. Major types are **Eastern Equine Encephalomyelitis (EEE)**, **Western Equine Encephalomyelitis (WEE)**, and **Venezuelan Equine Encephalomyelitis (VEE)**.

CLASSIFICATION

Baltimore Group IV. Family Togaviridae, Genus Alphavirus. ssRNA positive sense, enveloped, icosahedral nucleocapsid, 60–70 nm.

PROPERTIES

Enveloped with E1 and E2 glycoproteins. E2 is the main antigen for neutralizing antibodies. Sensitive to ether/detergents. Stable at pH 7–8.

VECTORS & CYCLE

VIRUS	PRIMARY VECTOR	RESERVOIR	DISTRIBUTION
EEE	<i>Culiseta melanura</i>	Wild birds	Eastern North America, Caribbean, South America
WEE	<i>Culex tarsalis</i>	Wild birds (passerines)	Western North America, South America
VEE	Multiple <i>Culex</i> , <i>Aedes</i>	Rodents (enzootic); horses (epizootic)	Central and South America

Horses and humans are **dead-end hosts** for EEE and WEE (don't develop sufficient viremia to infect mosquitoes). In epizootic VEE, horses develop high viremia and amplify the virus.

CLINICAL SIGNS

Fever, depression, impaired vision, circling, inability to swallow, head pressing, convulsions, coma. EEE has highest mortality (70–90% in horses). WEE mortality ~20–30%. VEE mortality variable.

DIAGNOSIS

Serology: HI, VN, ELISA. Paired samples (4× rise confirms recent infection). Virus isolation from brain tissue, blood. RT-PCR. Brain histology: encephalitis with perivascular lymphocytic infiltration and neuronal necrosis.

VACCINES

Inactivated vaccines available for EEE, WEE, and VEE. Combination equine vaccines common. Annual vaccination required. VEE MLV vaccine available for outbreak use. Vector control (mosquito reduction) important.

PUBLIC HEALTH

All three viruses are zoonotic. EEE has ~35% fatality in humans. Humans: dead-end hosts (EEE, WEE) but epizootic VEE can infect humans through biting arthropods.

Classical Swine Fever

Classical Swine Fever (CSF) — also called **Hog Cholera** — is a highly contagious, OIE List A viral hemorrhagic disease of pigs caused by **Classical Swine Fever Virus (CSFV)**.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group IV — Positive sense RNA
Family	Flaviviridae
Genus	Pestivirus
Species	Classical swine fever virus

MORPHOLOGY

Enveloped, icosahedral, 40–60 nm, ssRNA positive sense. Related to BVD virus and Border disease virus. Three major envelope glycoproteins: **E^{ns}**, **E1**, **E2**. E2 is the major protective antigen used in subunit vaccines.

STRAINS

One serotype, three genotypes. Genotype 1: American/European strains. Virulence varies from highly virulent (acute, fatal) to low virulence (chronic, subclinical).

HOSTS & TRANSMISSION

Pigs and wild boar only. Transmission: direct contact with infected animals, oral ingestion of contaminated feed/swill containing infected pork, iatrogenic, fomites. Transplacental transmission causes reproductive failure.

CLINICAL SIGNS

Acute form: High fever (41–42°C), conjunctivitis, cyanosis of skin (ears, abdomen, tail — **purple coloration**), huddling, haemorrhagic diarrhoea, neurological signs (ataxia, convulsions), high mortality (100% in naive herds). **Chronic form:** Persistent low-grade infection, growth retardation, intermittent fever. **Congenital form:** Persistently infected piglets (resembles BVD-PI).

PATHOLOGY

Button ulcers in large intestine (lymphoid follicle necrosis + haemorrhage — **pathognomonic**). Haemorrhagic lymph nodes, petechiae in kidneys, spleen infarcts, splenic marginal haemorrhages, encephalitis.

CSF BUTTON ULCERS — PATHOGNOMONIC

Circular, raised, necrotic, haemorrhagic lesions (button ulcers) in the large intestine at the level of lymphoid follicles (Peyer's patches) — pathognomonic for Classical Swine Fever.

DIAGNOSIS

Virus isolation in porcine kidney cells (SK6, CPK). FAT on tonsil/lymph node. ELISA, VN test. RT-PCR. Differentiate from African Swine Fever (ASF is not vaccine-preventable).

VACCINES

MLV vaccines (C strain — Chinese/Japanese origin; Thiverval strain). E2 subunit vaccine (allows DIVA — differentiating infected from vaccinated animals). Vaccination prohibited in disease-free countries — stamping out policy instead.

Bovine Viral Diarrhoea & Mucosal Disease

Bovine Viral Diarrhoea (BVD) and **Mucosal Disease (MD)** are caused by **Bovine Viral Diarrhoea Virus (BVDV)**. BVD is one of the most economically important diseases of cattle worldwide.

CLASSIFICATION

Baltimore Group IV. Family Flaviviridae, Genus Pestivirus. Two species: BVDV-1 and BVDV-2. Related to CSF virus and Border disease virus of sheep. Enveloped, 40–60 nm, ssRNA positive sense.

BIOTYPES

Two biotypes based on cell culture behavior: **Cytopathic (CP)** biotype — causes vacuolation and cell death in culture; **Non-cytopathic (NCP)** biotype — no CPE in culture. NCP biotype causes persistent infection (PI) in calves. CP biotype causes mucosal disease in PI animals.

PERSISTENTLY INFECTED (PI) ANIMALS

If cows are infected with NCP-BVDV during **first 120 days of gestation**, before immune competence develops, the foetus becomes persistently infected. PI animals: immunotolerant to the virus (don't produce antibodies), lifelong viremia, appear normal but shed huge quantities of virus. PI animals are the main reservoir for BVDV spread. When a PI animal encounters CP-BVDV (via superinfection or mutation) → **Mucosal Disease (MD)**.

MUCOSAL DISEASE

Occurs exclusively in PI animals when exposed to CP biotype. Acute, fatal disease: severe diarrhoea, oral erosions, laminitis, interdigital erosions. Mortality near 100%. No treatment effective.

CLINICAL SIGNS (ACUTE BVD)

Fever, leukopenia, nasal/ocular discharge, mild diarrhoea, oral erosions, drop in milk production. Reproductive: early embryonic death, abortion, congenital defects (cerebellar hypoplasia, cataracts, growth retardation). Immunosuppression → secondary infections.

DIAGNOSIS

Virus isolation in BT cells (bovine turbinate) or BEK cells. Antigen ELISA on ear notch (for PI detection in calves). VN test, ELISA for serology. PCR. PI detection: pooled serum ELISA, ear notch ELISA, immunohistochemistry.

VACCINES & CONTROL

MLV and inactivated vaccines. Control strategy: identify and eliminate PI animals through herd testing, vaccinate remaining animals. New Zealand, Scandinavia have successful PI-eradication programmes.

Avian Infectious Bronchitis

Avian Infectious Bronchitis (AIB) is a highly contagious respiratory disease of chickens caused by **Infectious Bronchitis Virus (IBV)** — the first coronavirus to be identified (1931).

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group IV — Positive sense RNA
Family	Coronaviridae
Genus	Gammacoronavirus
Species	Avian coronavirus (IBV)

MORPHOLOGY

Enveloped, pleomorphic, **120 nm**. Positive-sense ssRNA, **largest RNA genome** of any virus (~27–32 kb). Crown-like projections of the **Spike (S) protein** give it the corona (crown) appearance. Structural proteins: S (spike), E (envelope), M (membrane), N (nucleocapsid).

SEROTYPES

Many serotypes and variants due to frequent mutations and recombination. Massachusetts (Mass) serotype is the prototype. Major serotypes: Mass, Connecticut, Arkansas, Holte, Gray. New variants emerge constantly — vaccination with one serotype doesn't protect against all others.

HOSTS & TRANSMISSION

Chickens only (turkeys can also be infected). Highly contagious — aerosol, direct contact, fomites. Incubation 18–36 hours. Spread through flock very rapidly.

CLINICAL SIGNS

Respiratory: coughing, sneezing, rales, gasping — especially in young chicks. **Urogenital (nephropathogenic strains):** uremia, swollen kidneys with urate deposits. **Reproductive (laying hens):** drop in egg production, poor-quality eggs (soft-shelled, watery albumen, rough/misshapen shells), permanent damage to oviduct if infected early in life.

DIAGNOSIS

Virus isolation in embryonated eggs (CWE via amniotic/allantoic route → embryo dwarfing, curling, urate deposition — **dwarf curled embryo** is characteristic). HI test (difficult due to variant antigens), VN test, ELISA, RT-PCR.

VACCINES & CONTROL

Live attenuated Mass vaccines (H120, H52). Combination vaccines. Killed vaccines for layers before lay. Regular boosters needed due to new variants. Biosecurity measures essential.

Transmissible Gastroenteritis (TGE)

Transmissible Gastroenteritis (TGE) is an acute, highly contagious enteric disease of pigs caused by **Transmissible Gastroenteritis Virus (TGEV)**. Characterized by watery diarrhoea, vomiting, and high mortality in piglets under 2 weeks of age.

CLASSIFICATION

Baltimore Group IV. Family Coronaviridae, Genus Alphacoronavirus. Related to Porcine Epidemic Diarrhoea Virus (PEDV) and Porcine Respiratory Coronavirus (PRCV). Enveloped, ~120 nm, +ssRNA.

TRANSMISSION

Faecal-oral route. Contaminated faeces, feed, water, fomites. Starlings and other birds can mechanically transmit virus between farms.

CLINICAL SIGNS

Profuse watery/frothy diarrhoea, vomiting, rapid dehydration. Mortality near 100% in piglets <1 week. Older pigs survive but lose weight. Sows: loss of appetite, diarrhoea, reduced milk production (agalactia).

PATHOGENESIS

Virus destroys villous epithelial cells in small intestine → villous atrophy → malabsorptive diarrhoea. Young pigs have immature crypts unable to compensate for villous cell loss.

DIAGNOSIS

FAT or IHC on intestinal sections. ELISA, VN, AGID. RT-PCR for TGEV vs PRCV differentiation. Histology: villous atrophy.

| VACCINES & CONTROL

Live attenuated and inactivated vaccines. Vaccinate sows before farrowing — lactogenic immunity (IgA in colostrum/milk) protects piglets. All-in/all-out farrowing, disinfection. No treatment available; supportive electrolyte therapy for surviving pigs.

Bluetongue

Bluetongue (BT) is a non-contagious, insect-transmitted viral disease of ruminants caused by **Bluetongue Virus (BTV)**. OIE List A disease primarily affecting sheep; cattle are often subclinical but important as reservoir hosts.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group III — dsRNA
Family	Reoviridae
Genus	Orbivirus
Species	Bluetongue virus

MORPHOLOGY

Non-enveloped, icosahedral, **70–80 nm**. Genome: **10 segments of dsRNA**. Double-layered capsid. This segmentation enables reassortment. Stable at neutral pH. Ether-resistant (non-enveloped).

SEROTYPES

27 serotypes recognized. No cross-protection between serotypes. Serotypes 1, 2, 8, 10, 17 most prevalent. In India: serotypes 1, 2, 4, 9, 12, 16, 18, 23 reported.

VECTOR

Transmitted exclusively by biting midges of the genus **Culicoides** (mainly *C. imicola* in Africa/Europe, *C. variipennis* in North America). Not contagious between animals. Distribution follows vector distribution.

HOSTS

All ruminants. Sheep most severely affected. Cattle and goats usually subclinical. Cattle are important amplifying hosts (long viremia). White-tailed deer are severely affected in USA.

PATHOGENESIS

Culicoides bite → virus in bloodstream → replicates in endothelium and inflammatory cells → vasculitis → tissue edema, hemorrhage, necrosis. Tongue, coronary band, and muscle most affected.

CLINICAL SIGNS (SHEEP)

High fever, severe facial edema (especially nostrils), hyperemia and swelling of mouth mucosa, salivation, ulcerations on tongue/lips/nares, nasal discharge (later mucopurulent), **cyanotic blue tongue** (not always present — a misnomer!), coronary band inflammation → lameness, wool break. Mortality 2–30%; survivors may have prolonged convalescence.

DIAGNOSIS

Virus isolation in embryonated eggs (yolk sac), BHK-21 cells. AGID (group-specific), ELISA (competitive — serotype-specific). RT-PCR. VN test for serotyping.

VACCINES & CONTROL

Polyvalent MLV vaccines containing multiple serotypes. Inactivated vaccines safer for use in pregnant animals. Control: Vector (Culicoides) control, housing animals during peak biting times. Movement restrictions during outbreaks.

African Horse Sickness

African Horse Sickness (AHS) is a non-contagious, arthropod-borne disease of equids caused by **African Horse Sickness Virus (AHSV)**. OIE List A disease. Mortality up to 95% in naive horses.

CLASSIFICATION

Baltimore Group III. Family Reoviridae, Genus Orbivirus. Related to Bluetongue virus. Non-enveloped, 70 nm, 10 segments dsRNA. 9 serotypes.

VECTOR & DISTRIBUTION

Transmitted by **Culicoides imicola** and other Culicoides species. Endemic in sub-Saharan Africa. Periodic outbreaks in North Africa, Middle East, Spain, Portugal. Not present in Americas or Asia. Seasonal — follows Culicoides activity.

HOSTS

Horses most severely affected (95% mortality in acute form). Mules: ~50% mortality. Donkeys and zebras: usually subclinical (reservoir). Zebras are the natural host/reservoir.

CLINICAL FORMS

Pulmonary form (Dunkop — "thin lungs"): Most acute; high fever, severe dyspnea, paroxysmal coughing, frothy nasal discharge (foamy pulmonary edema). Death in hours. **Cardiac form (Dikkop — "thick head"):** Subcutaneous/supraorbital edema, edema of head/neck, petechiae on conjunctiva, death in 7–10 days. **Mixed form:** Combination of above — most common. **Horse sickness fever:** Mildest form; fever only, recovery usual.

DIAGNOSIS

Clinical signs + geographic location. Virus isolation in BHK-21 cells, suckling mice. ELISA (group and type-specific), VN test. RT-PCR for serotyping.

VACCINES & CONTROL

Polyvalent MLV vaccine (serotypes 1, 3, 4 in one and 2, 6, 7, 8 in another — given separately). Horses moved to disease-free areas before vector season. Strict controls on movement of equids from endemic areas.

Infectious Bursal Disease (Gumboro)

Infectious Bursal Disease (IBD) — also called **Gumboro disease** — is an acute, highly contagious immunosuppressive disease of young chickens caused by **Infectious Bursal Disease Virus (IBDV)**. Named after Gumboro, Delaware, USA, where first described in 1962.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group III — dsRNA
Family	Birnaviridae
Genus	Avibirnavirus
Species	Infectious bursal disease virus

MORPHOLOGY

Non-enveloped, icosahedral, 60 nm. Genome: **2 segments of dsRNA** (A and B). Segment A encodes VP2, VP3, VP4, VP5. **VP2** is the major protective antigen (serotype-specific). Segment B encodes VP1 (RNA polymerase).

SEROTYPES

2 serotypes: Serotype 1 (pathogenic in chickens, serotype 2 non-pathogenic). Very virulent IBDV (vvIBDV) causes more severe disease and high mortality. Antigenic variants evade vaccine-induced immunity.

TARGET ORGAN

The **Bursa of Fabricius** (cloacal bursa) — site of B lymphocyte development in birds. IBDV selectively destroys immature B lymphocytes in the bursa → profound immunosuppression → susceptibility to secondary infections and poor response to vaccination.

WHY BURSA MATTERS

Bursa of Fabricius is the avian equivalent of bone marrow for B-cell maturation. IBDV destroys it → B-cell immunosuppression → increased susceptibility to secondary infections (E. coli, Marek's disease, Newcastle disease). Most important in birds 3–6 weeks of age.

HOSTS & SUSCEPTIBILITY

Chickens (primarily 3–6 weeks of age). Turkeys subclinically infected. Very resistant to environmental inactivation — virus survives >100 days in litter and >52 days in water.

TRANSMISSION

Highly contagious. Faecal-oral route. Virus very stable in environment — difficult to eradicate from contaminated premises. Darkling beetles (*Alphitobius diaperinus*) serve as mechanical vectors within poultry houses.

CLINICAL SIGNS

In susceptible 3–6 week old chicks: sudden onset, depression, ruffled feathers, watery diarrhoea, vent pecking, prostration. Mortality varies 0–30% (up to 80–100% with vvIBDV). Recovered birds show immunosuppression. Bursa initially enlarged (oedematous) then atrophies rapidly.

PATHOLOGY

Bursa: initially enlarged, edematous, haemorrhagic; later atrophied. Petechial and ecchymotic haemorrhages in thigh and pectoral muscles. Kidneys pale, swollen. Histology: lymphocyte depletion in bursal follicles.

DIAGNOSIS

Gross lesions (bursal changes). Virus isolation in embryonated eggs (CAM inoculation) or cell culture. AGP test (agar gel precipitation). ELISA (most widely used). VN test for serotyping. RT-PCR.

VACCINES

Immune complex vaccines: Vaccine virus + specific antibody complex → delayed release, overcomes maternal antibodies. Live attenuated vaccines: 'hot', 'intermediate', 'mild' strains. Killed oil-emulsion

vaccines for breeders → high maternal antibody in progeny. Vector vaccines (turkey herpesvirus expressing VP2). Exact timing of live vaccine crucial — must overcome maternal antibodies without causing excessive immunosuppression.

Equine Infectious Anaemia

Equine Infectious Anaemia (EIA) — also called **Swamp fever** — is a persistent viral infection of horses caused by **EIA virus (EIAV)**. A Lentivirus that establishes lifelong infection.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group VI — ssRNA-RT (Retrovirus)
Family	Retroviridae
Subfamily	Orthoretrovirinae
Genus	Lentivirus

MORPHOLOGY

Enveloped, spherical, 80–100 nm. ssRNA (+sense), diploid genome. Reverse transcriptase (RT) converts viral RNA to DNA → integrates as provirus into host genome (permanent integration). Major proteins: Gag (MA, CA, NC), Pol (RT, IN, PR), Env (SU, TM).

REPLICATION — KEY FEATURE

RNA → (Reverse Transcriptase) → DNA → integrase → host genome (provirus) → permanent infection. The provirus is replicated as part of the host chromosome — **lifelong carrier state**.

ANTIGENIC VARIATION

EIAV undergoes extensive antigenic variation of its envelope glycoprotein (gp90) during persistent infection — the immune response selects virus variants that escape antibody neutralization. This leads to cyclical fever episodes.

CLINICAL SIGNS

Acute form: High fever (39.5–41°C), thrombocytopenia, anaemia, oedema of lower limbs and belly, weight loss, weakness. **Chronic form:** Repeated episodes of fever, weight loss, progressive anaemia, oedema. **Inapparent (carrier):** No clinical signs but lifelong viremia — most common.

DIAGNOSIS — COGGINS TEST

The **Coggins Test** (AGID — Agar Gel Immunodiffusion) is the gold standard serological test. Detects antibody to p26 (core antigen). ELISA (cELISA) is an alternative. Required for interstate and international movement of horses in USA and many countries. Infected horses must be euthanized or quarantined (minimum 200 meters from other equids).

TRANSMISSION

Mechanical transmission by biting flies (*Tabanus* spp. — horseflies, deerflies). Iatrogenic (contaminated needles, surgical equipment). Blood transfusion. Transplacental transmission.

VACCINES & CONTROL

No effective commercial vaccine available. EIAV's antigenic variability makes vaccination extremely difficult (similar challenge as HIV in humans). Control: Coggins testing before movement, quarantine/euthanasia of positive horses, insect control, strict hygiene.

Sheep Pox and Goat Pox

Sheep Pox and Goat Pox are caused by **Sheeppox virus (SPPV)** and **Goatpox virus (GTPV)** respectively. OIE List A diseases — the most severe of the poxvirus diseases of animals.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group I — dsDNA
Family	Poxviridae
Subfamily	Chordopoxvirinae
Genus	Capripoxvirus

MORPHOLOGY

Large, brick-shaped, enveloped, dsDNA, 200–300 nm. Replicates entirely in cytoplasm (unique among DNA viruses). Forms characteristic cytoplasmic inclusion bodies (**Bollinger bodies / eosinophilic inclusions**). Related to Lumpy Skin Disease virus (LSDV) in same genus.

HOSTS & STRAINS

Sheeppox virus: sheep primarily (some strains also infect goats). Goatpox virus: goats primarily (some strains infect sheep). Both in genus Capripoxvirus with LSDV. Many strains share serological cross-reactivity.

CLINICAL SIGNS

Fever, cutaneous lesions progressing through papule → macule → vesicle → pustule → scab/necrosis. Most severe on hairless skin (inner thighs, axillae, perineum, eyelids). **Typical pock lesion** = raised central necrotic eschar. Pulmonary form common in severe cases. Mortality up to 50% in naive flocks; higher in lambs. Characteristic stone-hard skin lesions in severe form.

DIAGNOSIS

Characteristic pock lesions + fever. Virus isolation in Vero cells or lamb kidney cells — CPE includes syncytia and intracytoplasmic inclusions. Electron microscopy (brick-shaped virion). AGID, VN test, ELISA. PCR for genus/species differentiation.

VACCINES

Live attenuated vaccines (Romania and Kenya strains). One dose provides >3 years immunity.

Heterologous protection: LSDV vaccine protects sheep and goats against sheeppox/goatpox (and vice versa) due to cross-immunity within Capripoxvirus.

Fowl Pox

Fowl Pox is a slow-spreading viral disease of birds caused by **Fowlpox virus (FWPV)**. Two clinical forms occur: cutaneous (dry) pox and diphtheritic (wet) pox.

CLASSIFICATION

Baltimore Group I. Family Poxviridae, Genus Avipoxvirus. Large dsDNA virus, brick-shaped, replicates in cytoplasm. Multiple avipoxvirus species including Fowlpox, Canarypox, Pigeonpox, Turkeypox.

CLINICAL FORMS

Cutaneous (dry) form: Proliferative nodular lesions (pocks) on unfeathered skin — comb, wattles, eyelids. Lesions progress papule → vesicle → pustule → scab. **Diphtheritic (wet) form:** Yellow-white caseous plaques on mucous membranes of mouth, pharynx, oesophagus, trachea — can cause asphyxiation.

BOLLINGER BODIES

Intracytoplasmic inclusion bodies within epithelial cells of pock lesions. Within Bollinger bodies are smaller elementary bodies called **Borrel bodies**.

TRANSMISSION

Mechanical transmission by biting mosquitoes (main vector in field), direct contact with infected birds, and contaminated feed/water. Virus persists in dried scabs for months to years.

DIAGNOSIS

Gross and histological lesions (Bollinger bodies). Virus isolation in embryonated eggs (pock lesions on CAM). Fluorescent antibody test.

VACCINES

Wing-web vaccination with MLV fowlpox vaccine from 4 weeks of age. Pigeonpox vaccine can also be used in chickens. Recombinant fowlpox vaccines (expressing genes from other pathogens — NDV, AI)

also available.

Contagious Ecthyma (Orf)

Contagious Ecthyma (Orf) — also called **Contagious Pustular Dermatitis** or Scabby Mouth — is a poxviral skin disease of sheep and goats caused by **Orf virus (ORFV)**. An important zoonosis.

CLASSIFICATION

Baltimore Group I. Family Poxviridae, Genus Parapoxvirus. Ovoid-shaped (not typical brick-shaped of orthopoxviruses), ~260 × 160 nm, dsDNA.

CLINICAL SIGNS

Proliferative papulovesicular lesions progressing to thick scabs on lips, nostrils, eyelids, oral mucosa, coronary band, teats of nursing ewes. Lesions can be severe enough to prevent feeding. Self-limiting in most cases (recovery 3–4 weeks). Secondary bacterial infection can complicate course.

ZOONOSIS

ORFV infects humans — **ecthyma contagiosum**. Single large nodular lesion on hand/arm, self-limiting (heals in 4–6 weeks). No specific treatment needed. Important in shepherds, slaughterhouse workers, veterinarians.

DIAGNOSIS

Clinical appearance. EM (parapoxvirus morphology). Virus isolation in lamb kidney cells or primary sheep cells. PCR. ELISA.

VACCINES

Live virus rubbed into scarified skin (leg of lamb). Scabs from lesions used as source. Important: Virus persists in dried scabs for years — avoid contaminating premises. No cross-immunity with other poxviruses.

Other Pox Diseases

Other important pox diseases of veterinary significance include **Lumpy Skin Disease (LSD)**, **Camelpox**, **Swinepox**, and **Buffalopox**.

| LUMPY SKIN DISEASE (LSD)

Caused by Lumpy Skin Disease Virus (LSDV) — a Capripoxvirus (same genus as Sheeppox/Goatpox). OIE List A disease. Transmitted by insects (Culicoides, mosquitoes, stable flies). Clinical signs: fever, cutaneous nodules (3–5 cm) over entire body, enlarged lymph nodes, edema of limbs. Diagnosis: virus isolation, PCR, ELISA. Vaccine: Neethling strain live attenuated vaccine; cross-protection with Sheeppox/Goatpox vaccines.

| CAMELPOX

Orthopoxvirus (genus Orthopoxvirus). Causes systemic or localized pock disease in camels. NOT zoonotic (unlike monkeypox or cowpox). Diagnosis: EM, PCR, AGID.

| SWINEPOX

Suipoxvirus (genus Suipoxvirus). Generalized cutaneous pock lesions in pigs. Transmitted by pig louse (*Haematopinus suis*) and direct contact. Usually self-limiting.

| BUFFALOPOX

Caused by Buffalopox virus (BPXV) — an Orthopoxvirus closely related to Vaccinia virus. Affects water buffaloes in India and Egypt. **Zoonotic** — infects humans in contact with infected buffaloes (milkers, veterinarians).

African Swine Fever

African Swine Fever (ASF) is a highly contagious hemorrhagic disease of domestic and wild pigs caused by **African Swine Fever Virus (ASFV)**. OIE List A disease. Currently a major pandemic threat globally with massive spread since 2018.

CLASSIFICATION

RANK	TAXON
Baltimore Group	Group I — dsDNA
Family	Asfarviridae
Genus	Asfivirus
Species	African swine fever virus

ASFV is the **only member of the family Asfarviridae** and the **only known DNA arbovirus** (transmitted by arthropods — soft ticks *Ornithodoros*).

MORPHOLOGY

Enveloped, icosahedral, **200 nm**. Large dsDNA genome (~170–190 kb). Complex multi-layered structure. Replicates in cytoplasm in perinuclear "viral factories." Approximately 150 open reading frames — about 50 structural proteins.

RESERVOIR & EPIDEMIOLOGY

In Africa: **Sylvatic cycle** — Warthogs (asymptomatic) → *Ornithodoros* ticks → Warthogs (lifelong). Domestic pig cycle in endemic countries. Trans-boundary spread via trade in infected animals, contaminated pork products, and infected ticks.

CLINICAL SIGNS

Peracute: Sudden death with minimal signs. **Acute:** Fever (42°C), anorexia, cyanosis of skin (ears, snout, limbs — turns blue/purple), haemorrhagic diarrhoea, abortion, ataxia, death within 7–10 days. Mortality near 100% in highly virulent strains. **Subacute/Chronic:** With moderately virulent strains — respiratory signs, weight loss, skin ulcers, joint swelling.

PATHOLOGY

Haemorrhagic lymph nodes (especially renal and gastrointestinal), splenomegaly (massively enlarged, haemorrhagic spleen), petechiae throughout body. Histology: haemorrhage and necrosis in lymphoid tissue.

DIAGNOSIS

No simple field test available. ELISA (antigen + antibody), IFAT, PCR (most sensitive). Haemadsorption test (HAD) — ASFV causes haemadsorption of pig red blood cells on infected macrophages in culture — **pathognomonic in culture**. Counter immunoelectrophoresis (CIE). Differentiate from CSF — both cause hemorrhagic disease in pigs.

ASF VS CSF — KEY DIFFERENCE

Both cause haemorrhagic disease in pigs. **CSF:** caused by Pestivirus (RNA virus) — preventable by vaccination. **ASF:** caused by Asfivirus (DNA virus) — NO vaccine available.

Haemadsorption test positive only for ASF.

VACCINES & CONTROL

NO vaccines available. Control: strict biosecurity, movement controls, stamping-out. No specific treatment. Contaminated pork/pork products are the main vehicle for international spread. All food waste must be properly treated before feeding to pigs.

Herpesviruses

Herpesviruses (Family Herpesviridae) are large, enveloped dsDNA viruses that cause **latent and recurrent infections**. The family has ~100 members infecting a wide range of host species. Name from Greek "herpein" = to creep, reflecting their characteristic recurrent nature.

CLASSIFICATION & SUBFAMILIES

SUBFAMILY	CHARACTERISTICS	VETERINARY MEMBERS
Alphaherpesvirinae	Fast replicating, neurotropic, latency in sensory ganglia	PRV, IBR/BHV-1, EHV-1/4, FHV-1, MDV, HSV
Betaherpesvirinae	Slow replicating, large cell inclusion bodies, latency in monocytes	CMV, HHV-6
Gammaherpesvirinae	Lymphotropic, latency in B/T lymphocytes	Marek's disease-like

MORPHOLOGY (COMMON TO ALL)

Enveloped, 150–200 nm. dsDNA linear genome. Four components: (1) Core (dsDNA + protein), (2) Icosahedral capsid, (3) Tegument (protein layer between capsid and envelope), (4) Lipid envelope with glycoprotein spikes.

REPLICATION

Glycoproteins mediate attachment → entry by fusion. DNA released into nucleus → immediate-early (IE) → early (E) → late (L) gene expression. DNA replication in nucleus. Assembly in nucleus → envelopment by nuclear membrane budding → release by exocytosis. Latency: circular episomal DNA in nucleus, minimal gene expression.

| IMPORTANT VETERINARY HERPESVIRUSES

1. PSEUDORABIES (AUJESZKY'S DISEASE) — PRV/SUV-1

Affects pigs (reservoir, mild disease), but fatal neurological disease in cattle, sheep, goats, dogs, cats.

Name is misleading — not related to rabies. **Intense pruritus** (pruritis unlike true rabies) at bite site in non-porcine species — called "mad itch." Cultivated in PK-15 cells (pig kidney). Controlled in Europe by MLV and gE-deleted marker vaccines (DIVA strategy). OIE notifiable.

2. INFECTIOUS BOVINE RHINOTRACHEITIS — IBR/BHV-1

Caused by Bovine Herpesvirus-1 (BHV-1 / BoHV-1). Multiple clinical syndromes: IBR (respiratory), IPV (infectious pustular vulvovaginitis/balanoposthitis), encephalitis, abortion. Latency in trigeminal ganglia. Reactivated by stress/corticosteroids. Diagnosis: virus isolation, VN, ELISA. gE-deleted marker vaccines widely used in Europe for IBR eradication programmes.

3. EQUINE HERPESVIRUS 1 & 4 — EHV-1, EHV-4

EHV-1: Abortion in mares (3rd trimester), neonatal death, neurological disease (EHM — equine herpesvirus myeloencephalopathy). EHV-4: Rhinopneumonitis (respiratory disease). Latency in trigeminal ganglia and leukocytes. Transmitted by aerosol and nasal discharge. Vaccines available (killed and MLV).

4. MAREK'S DISEASE VIRUS — MDV/GAHV-2

Gammaherpesvirus. Causes lymphoproliferative disease in chickens: T-cell lymphoma affecting peripheral nerves (paralysis — "grey eye", "range paralysis"), visceral tumors. Transmitted via feather follicle epithelium in dander. Vaccine: Turkey Herpesvirus (HVT) — first vaccine to prevent cancer. Bivalent (HVT + SB-1) and trivalent vaccines available. MDV shows evolution towards increased virulence despite vaccination (very virulent plus strains).

5. FELINE HERPESVIRUS 1 — FHV-1

Causes feline viral rhinotracheitis (FVR) — major upper respiratory disease of cats. Signs: sneezing, nasal/ocular discharge, conjunctivitis, keratitis, corneal ulcers. Lifelong latency with recrudescence during stress. Part of FVRCP vaccine.

6. INFECTIOUS LARYNGOTRACHEITIS — ILTV/GAHV-1

Alphaherpesvirus of chickens. Severe respiratory disease with haemorrhagic exudate in trachea — characteristic "blood stained" trachea. High mortality. Diagnosis: histology (intranuclear inclusions in tracheal cells), virus isolation, PCR. MLV vaccines (chicken embryo origin, tissue culture origin).

Adenoviruses

Adenoviruses (Family Adenoviridae) are non-enveloped dsDNA viruses that cause a range of respiratory, enteric, and hepatic diseases in various animal species.

CLASSIFICATION

Baltimore Group I. Family Adenoviridae, four genera: Mastadenovirus (mammals), Aviadenovirus (birds), Atadenovirus (various), Siadenovirus (frogs, turkeys). Non-enveloped, icosahedral, 70–90 nm. dsDNA linear genome.

MORPHOLOGY

Capsid made of 252 capsomeres: 240 hexons + 12 pentons. **Fibers** project from each penton — responsible for hemagglutination and cell attachment. Hexon is the major type-specific antigen. The name comes from "adenoid tissue" where first isolated (human tonsils, 1953).

IMPORTANT VETERINARY ADENOVIRUSES

1. INFECTIOUS CANINE HEPATITIS (ICH) — CAdV-1

Canine Adenovirus type 1 (CAdV-1). Acute hepatitis in dogs, foxes, wolves. Rubarth's disease. Clinical signs: sudden death (peracute), fever, hepatomegaly, abdominal pain, haemorrhagic diarrhoea, corneal edema (**"blue eye"** — immune complex deposition in cornea during recovery). Intranuclear basophilic inclusion bodies in hepatocytes. Vaccine: CAdV-2 MLV (heterologous — avoids "blue eye" complication of CAdV-1 vaccine).

2. EQUINE ADENOVIRUS

EAdV-1 and EAdV-2. Respiratory disease in foals, especially Arabian foals with combined immunodeficiency (SCID) — fatal pneumonia. Diagnosis: virus isolation, paired serology.

3. BOVINE ADENOVIRUSES (BADV)

Multiple types (BADV-1 to -10). Cause respiratory, enteric, conjunctival, and urogenital diseases in cattle. BADV-3 most commonly isolated. Usually subclinical or mild unless combined with other pathogens.

4. FOWL ADENOVIRUSES

Multiple species in Aviadenovirus. Inclusion Body Hepatitis (IBH): acute hepatitis with intranuclear inclusions in chickens (FAdV-4, FAdV-11). Hydropericardium Syndrome (HPS / Angara disease): pericardial effusion, hepatitis, high mortality in broilers — caused by FAdV-4 (HB1/Pakistan/Angara strains). Egg Drop Syndrome 1976 (EDS-76): caused by Duck Adenovirus 1 — affects layer chickens, causes drop in egg production, thin/shell-less eggs. Atadenovirus (Duck adenovirus).

EGG DROP SYNDROME (EDS-76)

Caused by Duck Adenovirus 1 (DAdV-1 / EDS virus). Affects chickens — not ducks. Causes sudden drop in egg production, thin-shelled/shell-less/soft-shelled eggs, pale/brown eggs. Vaccine available (inactivated).

Parvoviruses

Parvoviruses (Family Parvoviridae) are among the smallest DNA viruses. Non-enveloped, single-stranded, linear DNA. Despite their simple structure, they cause serious diseases in animals.

CLASSIFICATION

Baltimore Group II — ssDNA. Family Parvoviridae, Subfamily Parvovirinae (vertebrate hosts). Non-enveloped, icosahedral, 18–26 nm. ssDNA genome. Requires dividing cells for replication (uses host DNA polymerase for dsDNA intermediate).

IMPORTANT VETERINARY PARVOVIRUSES

1. CANINE PARVOVIRUS — CPV-2

Emerged around 1978 (CPV-2) — derived from Feline panleukopenia virus. Now variants CPV-2a, 2b, 2c. Two clinical syndromes: **Enteritis form** (most common) — acute haemorrhagic enteritis in puppies, severe leukopenia, vomiting, bloody diarrhoea, high mortality; **Myocarditis form** (now rare) — myocardial necrosis in pups < 8 weeks. Highly resistant — survives months in environment. Diagnosis: ELISA (antigen detection in faeces), PCR. Vaccines: MLV vaccines (CPV-2 based) protect against all variants including 2c.

2. FELINE PANLEUKOPENIA VIRUS — FPV

Also called feline infectious enteritis, feline distemper. Destroys rapidly dividing cells: intestinal crypts, bone marrow (→ panleukopenia), cerebellum (in utero → cerebellar hypoplasia). Signs: fever, vomiting, diarrhoea, collapse. Very high mortality in kittens. Vaccine: MLV FPV vaccine (part of FVRCP).

3. PORCINE PARVOVIRUS — PPV

Major cause of reproductive failure in pigs. **SMEDI syndrome**: Stillbirth, Mummification, Embryonic Death, Infertility. Virus crosses placenta in first trimester → kills embryos/foetuses at different stages. Sows immune, no clinical signs. Diagnosis: FA on foetal tissues, HI test. Vaccines: killed vaccines for gilts before breeding.

Papovaviruses

Papovaviruses are small, non-enveloped, circular dsDNA viruses divided into two families:

Papillomaviridae (papillomaviruses) and **Polyomaviridae** (polyomaviruses). The name "Papovavirus" was derived from Papilloma + Polyoma + Vacuolating agent (SV40).

CLASSIFICATION

Baltimore Group I. dsDNA (circular, supercoiled). Non-enveloped, icosahedral, 45–55 nm (papillomaviruses) and 40–45 nm (polyomaviruses). Cannot be propagated in conventional cell cultures — papillomaviruses require differentiating keratinocytes.

PAPILLOMAVIRUSES — WARTS/PAPILLOMAS

Cause benign epithelial tumors (warts/papillomas) in various animal species. **Bovine Papillomavirus (BPV-1 to -6)**: Cutaneous warts, fibropapillomas. BPV-1 associated with equine sarcoid (most common equine skin tumor). **Canine oral papillomatosis**: Multiple oral warts in young dogs — transmitted by direct contact. Self-limiting (regress spontaneously). **Equine sarcoid**: Not a true papilloma — locally aggressive fibroblastic tumor associated with BPV-1/2 (bovine papillomavirus in equine). Diagnosis: PCR, histology.

POLYOMAVIRUSES

Can cause oncogenic transformation in lab animals. SV40 (Simian Virus 40): used extensively in research — discovered as contaminant of early poliovirus vaccines. Murine polyomavirus: laboratory mouse tumor model. **Budgerigar fledgling disease**: caused by avian polyomavirus (APV) — causes hepatitis, hydrops, feather abnormalities in young budgerigars and other psittacines.

Circoviruses

Circoviruses (Family Circoviridae) are the **smallest known DNA viruses** of vertebrates. Single-stranded circular DNA, non-enveloped. Of major importance in pigs and birds.

CLASSIFICATION

Baltimore Group II. Family Circoviridae, Genus Circovirus and Cyclovirus. Non-enveloped, icosahedral, **17–22 nm**. Circular ssDNA genome (~1.7–2 kb) — smallest DNA virus. Replication requires S-phase of cell division.

PORCINE CIRCOVIRUS DISEASE

PORCINE CIRCOVIRUS TYPE 2 (PCV-2)

PCV-1: Non-pathogenic. PCV-2: Pathogenic — associated with multiple disease syndromes collectively called **Porcine Circovirus Associated Diseases (PCVAD)**:

- **Postweaning Multisystemic Wasting Syndrome (PMWS)**: Most important — progressive weight loss, respiratory distress, jaundice, lymphadenopathy in weaned pigs (6–15 weeks). High morbidity and mortality.
- **Porcine Dermatitis and Nephropathy Syndrome (PDNS)**: Skin lesions (raised red/brown papules), renal failure.
- **Porcine Respiratory Disease Complex (PRDC)**: PCV-2 acts as one component.
- **Reproductive failure**: PCV-2 causes myocarditis in aborted/stillborn piglets.

CHICKEN ANEMIA VIRUS (CAV)

Chicken Anemia Virus (Gyrovirus genus): causes aplastic anaemia, immunosuppression, and thymic atrophy in young chicks. Clinical signs: anaemia, haemorrhage, severe immunosuppression. Transmitted vertically (transovarian) and horizontally. Diagnosis: PCR, ELISA, FAT on thymus. Vaccines: MLV vaccines for breeders to prevent vertical transmission.

BEAK AND FEATHER DISEASE

Psittacine Beak and Feather Disease Virus (BFDV): Circovirus affecting parrots and cockatoos. Causes feather loss, beak deformity, immunosuppression. Currently no effective vaccine.

CONTROL OF PCV-2

Commercial subunit and killed whole-virus vaccines (ORF2 protein — major capsid protein). Vaccines dramatically reduce PMWS and improve production parameters. Biosecurity, stable management, reduce stress and co-infections.

⚡ MASTER QUICK REVISION — 60 MUST-KNOW POINTS

1. **Newcastle Disease** = Avian Paramyxovirus type 1 (APMV-1); Paramyxoviridae; Avulavirus; Ranikhet disease; OIE List A
2. **ND Pathotypes**: Velogenic (Hertz 33, Texas GB) → Mesogenic (K, R2B) → Lentogenic (LaSota, F) → Avirulent (V4); ICPI ≥ 0.7 = notifiable
3. **Rinderpest** = Morbillivirus; FIRST animal disease globally eradicated (June 2011 by FAO/OIE); Plowright vaccine (RBOK strain) achieved eradication
4. **PPR** = goat plague; Morbillivirus; first described 1942 West Africa; goats > sheep; AR-87 (Arasur strain) vaccine — Madras Veterinary College; OIE List A
5. **Canine Distemper** = Morbillivirus; hard pad disease (hyperkeratosis); Old Dog Encephalitis; Lentz bodies (intranuclear + intracytoplasmic inclusions)
6. **Avian Influenza** = Orthomyxoviridae; segmented genome (8 segments); HA (16 subtypes) + NA (9 subtypes); HPAI = H5/H7; OIE List A; pigs = mixing vessel
7. **Antigenic Drift** = point mutations in HA/NA (gradual); **Antigenic Shift** = reassortment of gene segments (sudden, pandemic risk)
8. **Equine Influenza** = H7N7 (Equi-1) and H3N8 (Equi-2); H3N8 now dominant; harsh dry cough characteristic; annual vaccination needed
9. **Swine Influenza** = subtypes H1N1, H3N2, H1N2; pigs have BOTH α -2,3 and α -2,6 sialic acid receptors → mixing vessel; 2009 pandemic H1N1 swine-origin
10. **Rabies** = Rhabdoviridae; Lyssavirus; bullet-shaped (180 × 75 nm); 5 proteins N, P, M, G, L; G-protein = neutralizing antibody target

11. **Negri bodies:** intracytoplasmic eosinophilic inclusions in neurons; best in hippocampus + cerebellum; Sellers stain = magenta with blue interior granules; pathognomonic for rabies
12. **Street virus:** from rabid animals, variable incubation, forms Negri bodies; **Fixed virus:** lab-adapted, short incubation, no Negri bodies, used for vaccines
13. **Rabies PEP** = wound cleaning + RIG (passive) + cell culture vaccine (HDCV/PCEC/PVRV); Semple vaccine (neural) obsolete; 4.6/1000 neuromuscular reactions
14. **BEF** = Bovine Ephemeral Fever; Rhabdoviridae, Ephemerovirus; transmitted by Culicoides; three-day sickness; not contagious; NSAIDs for treatment
15. **FMD** = 7 serotypes: O, A, C, Asia 1, SAT 1, SAT 2, SAT 3; NO cross-immunity between serotypes; OIE List A; non-enveloped Picornavirus (Aphthovirus)
16. **FMD VP1** protein = major antigenic site (RGD motif); pigs = amplifying host; cattle carrier up to 3.5 years; sheep 9 months; pigs do NOT become carriers
17. **FMD diagnosis:** NSP ELISA differentiates vaccinated from infected animals (non-structural protein antibodies); probe = vesicle epithelium
18. **Duck Viral Hepatitis** = Picornaviridae, Avihepatovirus; <3 weeks ducklings; opisthotonus (backward neck arching); haemorrhagic hepatitis; 95% mortality in <1 week
19. **Equine Encephalomyelitis** = Togaviridae, Alphavirus; EEE (Culiseta melanura), WEE (Culex tarsalis), VEE; horses + humans = dead-end hosts; EEE highest mortality (70–90%)
20. **Classical Swine Fever** = Hog Cholera; Flaviviridae, Pestivirus; button ulcers in large intestine (PATHOGNOMONIC); cyanosis of ears/abdomen/tail; OIE List A
21. **BVD-MD** = Bovine Viral Diarrhoea; Pestivirus; PI animals (persistent infection) = infected before day 120 of gestation; PI animal + CP biotype = Mucosal Disease (fatal)
22. **Avian Infectious Bronchitis** = first coronavirus described (1931); largest RNA genome; Mass serotype prototype; dwarf curled embryo in eggs; multiple serotypes — no cross-immunity
23. **TGE** = Transmissible Gastroenteritis; Coronaviridae, Alphacoronavirus; pigs; 100% mortality in <1 week piglets; villous atrophy; sow vaccination → lactogenic IgA
24. **Bluetongue** = Reoviridae, Orbivirus; 27 serotypes; 10 dsRNA segments; vector = Culicoides only; sheep most severe; blue tongue is RARE sign (misnomer!); OIE List A

25. **African Horse Sickness** = Orbivirus; 9 serotypes; Culicoides vector; Dunkop (pulmonary), Dikkop (cardiac = supraorbital oedema); zebra = reservoir; OIE List A; mortality up to 95%
26. **IBD / Gumboro** = Birnaviridae, Avibirnavirus; 2 dsRNA segments; VP2 = major protective antigen; destroys Bursa of Fabricius → B-cell immunosuppression; 3–6 weeks chickens
27. **Equine Infectious Anaemia** = Retroviridae, Lentivirus; lifelong infection; Coggins Test (AGID p26 antigen) = gold standard; swamp fever; NO effective vaccine
28. **Sheep Pox / Goat Pox** = Capripoxvirus; OIE List A; Bollinger bodies (cytoplasmic inclusions); brick-shaped pox virion; cross-immunity with LSD; one dose = 3 years immunity
29. **Fowl Pox** = Avipoxvirus; Bollinger bodies (containing Borrel bodies); two forms: cutaneous (dry) + diphtheritic (wet); wing-web vaccination; mosquito = vector
30. **Orf (Contagious Ecthyma)** = Parapoxvirus; sheep/goats; scabby mouth; ZOONOSIS (single nodule on hand, self-limiting); live virus vaccine; no cross-immunity with other poxviruses
31. **African Swine Fever** = Asfarviridae; ONLY DNA arbovirus; Ornithodoros tick = vector; haemadsorption test = pathognomonic; NO vaccine; cyanosis + massive splenomegaly + haemorrhagic LN
32. **Herpesviruses**: 3 subfamilies — Alpha (fast, neurotropic, latency in ganglia), Beta (slow, large inclusions), Gamma (lymphotropic); tegument between capsid and envelope
33. **Pseudorabies (PRV)** = intense pruritus (mad itch) in cattle/dogs/cats; pigs are reservoir; PK-15 cells; gE-deleted marker vaccines; OIE notifiable
34. **IBR (BHV-1)** = Bovine Herpesvirus-1; respiratory + IPV + abortion; latency in trigeminal ganglia; corticosteroids reactivate; gE-deleted DIVA vaccines for eradication
35. **Marek's Disease** = Gammaherpesvirus; T-cell lymphoma; peripheral nerve paralysis; HVT = FIRST vaccine to prevent cancer; feather follicle = source of infectious virus
36. **Canine Parvovirus (CPV-2)** = emerged ~1978 from FPV; haemorrhagic enteritis + panleukopenia in puppies; very stable virus; MLV vaccines effective against all variants
37. **Feline Panleukopenia** = FPV; cerebellar hypoplasia (in utero); panleukopenia; intestinal crypt destruction; part of FVRCP vaccine

38. **Porcine Parvovirus (PPV)** = SMEDI syndrome (Stillbirth, Mummification, Embryonic Death, Infertility); sows asymptomatic; killed vaccines for gilts before breeding
39. **Adenoviruses** = 252 capsomeres (240 hexons + 12 pentons); fibers from pentons for attachment; hexon = type-specific antigen; non-enveloped; dsDNA
40. **Infectious Canine Hepatitis (ICH)** = CAdV-1; blue eye (corneal oedema on recovery); Rubarth's disease; CAdV-2 vaccine used to avoid blue eye complication
41. **Hydropericardium Syndrome** = FAdV-4; Angara disease; broiler chickens India/Pakistan; pericardial effusion + hepatitis; high mortality
42. **EDS-76** = Egg Drop Syndrome 1976; Duck Adenovirus 1; affects CHICKENS (not ducks); soft/shell-less eggs; drop in production; killed vaccine
43. **Papovaviruses**: Papilloma (warts, circular dsDNA, 55 nm) + Polyoma (oncogenic potential); BPV-1 → equine sarcoid; SV40 discovered as polio vaccine contaminant
44. **Circoviruses** = SMALLEST vertebrate DNA viruses (17–22 nm); circular ssDNA; PCV-2 → PMWS + PDNS; CAV → aplastic anaemia in chicks; BFDV → beak and feather disease
45. **PMWS** = Postweaning Multisystemic Wasting Syndrome; PCV-2; wasting + lymphadenopathy + respiratory distress in 6–15 week pigs; subunit vaccine (ORF2 capsid protein)
46. **OIE List A diseases covered**: FMD, ND, HPAI, Rinderpest, CSF, ASF, AHS, Bluetongue, PPR, Sheep Pox/Goat Pox — all transboundary, rapid spread, severe socioeconomic impact
47. **Zoonotic viruses in this course**: Rabies, Avian Influenza (H5N1), Swine Influenza (H1N1 2009), Equine Encephalomyelitis, Orf (contagious ecthyma), Buffalopox, Newcastle Disease (conjunctivitis only)
48. **Insect-transmitted (Arboviruses)**: BT (Culicoides), AHS (Culicoides), EEE/WEE/VEE (mosquitoes), BEF (Culicoides/mosquitoes), ASF (Ornithodoros ticks); rabies = NOT arbovirus
49. **Carrier states**: FMD — cattle 3.5 years, sheep 9 months, pigs NO; BVD — PI animals (lifelong); EIA — lifelong provirus (Lentivirus); Herpesviruses — ganglionic latency
50. **Haemadsorption test** = diagnostic for ASF; Adsorption of pig RBCs to ASFV-infected macrophages in culture; not seen in CSF (key differential diagnosis)
51. **Coggins Test** = AGID test for EIA; detects antibody to p26 core protein; required for horse movement internationally; +ve horse must be quarantined or euthanized

52. **Immune complex IBD vaccines:**
Vaccine virus complexed with specific antibody → delayed release → overcomes maternal antibodies in commercial broilers
53. **DIVA strategy:** Differentiating Infected from Vaccinated Animals; used for IBR (gE-deleted vaccine), FMD (NSP ELISA), CSF (E2 subunit vaccine) — allows serological surveillance while vaccinating
54. **Largest RNA genome:** Coronaviruses (~27–32 kb); Smallest DNA virus: Circoviruses (17–22 nm); Bullet-shaped: Rhabdoviruses; Brick-shaped: Poxviruses
55. **Dwarf curled embryo** = IBV (Infectious Bronchitis Virus) in embryonated eggs; Tiger heart (pale myocardial streaks) = FMD in piglets; Button ulcers = CSF
56. **Bursa of Fabricius:** avian B-lymphocyte development organ; targeted by IBDV; absent/atrophied = severe B-cell immunosuppression; analogous to bone marrow B-cell development in mammals
57. **Fixed vs Street virus (Rabies):** Fixed = lab-adapted (short incubation, no Negri bodies, used in vaccines); Street = natural (long incubation, Negri bodies, neurotropic)
58. **Marek's Disease vaccine** = HVT (Herpesvirus of Turkeys) = world's FIRST vaccine to prevent cancer; MDV spreads via feather follicle dander — vaccine does not prevent infection, only disease
59. **Seven FMDV serotypes** = O, A, C, Asia 1, SAT 1, SAT 2, SAT 3; Only 3 serotypes in India (O, A, Asia 1); NSP ELISA to distinguish vaccinated from infected
60. **Poxvirus features:** Replicates entirely in cytoplasm (unlike other DNA viruses); brick-shaped; Bollinger bodies (Fowlpox), eosinophilic inclusions (Capripox); largest animal viruses
61. **Baltimore Group summary:** I=dsDNA (Herpes, Pox, Adeno, Asfa), II=ssDNA (Parvo, Circo, Papova), III=dsRNA (Reo, Birna), IV=+ssRNA (Picorna, Toga, Flavi, Corona), V=-ssRNA (Paramyxo, Orthomyxo, Rhabdo), VI=ssRNA-RT (Retro)